ITA0610-Machine Learning

LAB EXPERIMENTS

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Lab 14: Naïve Bayes Classification – Iris Dataset

Aim

To implement the **Naïve Bayes classification algorithm** using the Iris dataset.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Create a Gaussian Naïve Bayes classifier.
5. Train the model using the training data.
6. Predict the classes for the test data.
7. Calculate and display the accuracy.

Program

```
# Lab 14: Naïve Bayes Classification – Iris Dataset


from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score


# Load Iris dataset

iris = load_iris()


X = iris.data

y = iris.target


# Split the dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create and train the model
model = GaussianNB()
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

print("Predictions:", y_pred)
print("Actual Values:", y_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Sample Output

```
Predictions: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 1 0 2 2 2 2 2 0 0]
Actual Values: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 1 0 2 2 2 2 2 0 0]
Accuracy: 1.0
```

Result

Thus, the **Gaussian Naïve Bayes classifier was successfully implemented** using the Iris dataset, and the classes were predicted successfully.